

RadarMind-DriveLM v0.43: Transferring InternVL4Drive Insights with a Graph-Anchor Ensemble

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Abstract

We study the winning InternVL4Drive system from the CVPR 2024 Driving with Language track and transfer its reproducible insights to an existing Qwen2.5-VL DriveLM pipeline. InternVL4Drive combines six-view spatial layout, dynamic high resolution, box-oriented grounding, and complementary model ensembling. Our controlled experiment targets the last principle without changing training data or selecting answers using development labels. A fixed node router uses the Graph-SFT model for the first important-object anchor of each frame and the Mixture-of-LoRA (MoL) model for all downstream questions. On a 3,355-QA scene-isolated development split, the local DriveLM-DS proxy rises from 0.608245 to **0.612293**. On the exact 1,848-ID graph-eligible intersection, it rises from 0.609563 to **0.610557**, while Planning, Accuracy, Match, and multiple-choice accuracy remain unchanged. All semantic scores in this experiment are read from an existing SQLite cache by a network-disabled evaluator.

1 Motivation

DriveLM evaluates graph-structured visual question answering over synchronized surround-view images. The first important-object answer determines which object-dependent questions are eligible for downstream evaluation. This creates an asymmetric bottleneck: a model can be strong at Planning but lose coverage because its anchor coordinates are inaccurate.

The InternVL4Drive technical report explicitly exploits this property. It uses InternVL-1.5, arranges six views into a labeled 2×3 panorama, applies dynamic high-resolution tiling, converts point annotations to boxes using SAM, and observes that complementary v1/v2 models can be ensembled for a higher score. Its best single model reports 0.6002 on the official leaderboard.

Our prior experiments expose a complementary pair: MoL-700 is strongest in downstream Planning, whereas Graph-A improves anchor grounding and graph coverage but degrades same-ID Planning. This motivates node-level ensembling.

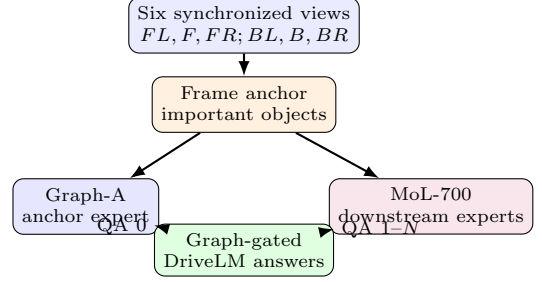


Figure 1: Fixed node-level routing transferred from the complementary ensemble observation in InternVL4Drive. Routing never reads a candidate answer or label.

2 Transferred Design

2.1 What we retain from InternVL4Drive

Figure 1 separates ideas that can be transferred immediately from those requiring a new training run. We retain fixed model ensembling and the view that anchor perception deserves a specialized representation. Mosaic input, explicit topology prompts, and box supervision are registered as future controlled ablations; directly changing a six-image-trained model to one mosaic at inference would introduce an avoidable distribution shift.

2.2 Fixed node router

For record i , the source is

$$s_i = \begin{cases} \text{Graph-A,} & \text{qa_index}_i = 0, \\ \text{MoL-700,} & \text{otherwise.} \end{cases} \quad (1)$$

The rule routes exactly 467 frame anchors to Graph-A and 2,888 downstream QA to MoL. It is independent of reference answers, candidate text, confidence, and development score. Both source files have 3,355/3,355 coverage.

2.3 Safety of semantic evaluation

The local evaluator follows the public DriveLM structure,

$$F = .4J + .2L + .2M + .2A, \quad (2)$$

where J is semantic Planning score, L combines BLEU-1-4, ROUGE-L and CIDEr, M combines coordinate F1 and

Table 1: Complete 3,355-QA DriveLM-DS evaluation.

Metric	MoL-700	v0.43	Δ
Eligible	1,911	1,927	+16
Accuracy	0.808735	0.813154	+0.004419
Judge /100	72.4769	73.0197	+0.5429
BLEU-1	0.772320	0.789634	+0.017315
BLEU-2	0.709429	0.724770	+0.015341
BLEU-3	0.651638	0.665099	+0.013461
BLEU-4	0.595859	0.607520	+0.011661
ROUGE-L	0.723957	0.724571	+0.000613
CIDEr	0.200502	0.198875	-0.001627
Match /100	30.7513	30.7513	0
Final	0.608245	0.612293	+0.004048

graph semantics, and A is exact accuracy. DeepSeek-V4-Flash substitutes for the unavailable official GPT judge, so we call the metric DriveLM-DS.

No new text is sent to an API. We implement a cache-only evaluator that does not construct a network client, opens SQLite in read-only mode, reproduces the frozen cache key, and fails on any miss. The promoted candidate obtains 791/791 cache hits and zero failures.

3 Experiments

3.1 Setup

The baseline is Qwen2.5-VL-7B with four task-specific rank-8 LoRA experts, selected at MoL step 700. Graph-A is a shared Graph-SFT adapter trained for 600 updates over complete Perception–Prediction–Planning–Behavior frame trajectories. Evaluation uses 3,355 scene-isolated QA. No model is retrained in this experiment, making the attribution to routing exact.

We test two predeclared rules: (A) Graph-A only for the frame anchor; and (B) Graph-A for the anchor and tag-3 grounding. Rule B has 132 semantic cache misses and is therefore excluded rather than assigned an incomplete Final.

3.2 Full-protocol result

Table 1 shows that the anchor ensemble increases eligible QA by 16, improves Accuracy by 0.44 percentage points and semantic Planning by 0.54 points, and raises Final by 0.004048 (0.67% relative). CIDEr slightly declines, which is explicitly retained in the report.

3.3 Same-ID audit

Candidate-dependent graph gating can make raw Finals incomparable. We therefore intersect the eligible IDs, yielding 1,848 common QA (633 tag-0, 616 tag-1, 467 tag-2, 132 tag-3). As Table 2 shows, every downstream metric is identical because those answers are exactly MoL outputs. Only Language changes, and Final still improves by 0.000993. All frozen promotion gates pass.

The paired lexical Token-F1 decreases by 0.000969 and a separate token-level ROUGE-L F1 decreases by 0.000474, although the official COCO Language aggreg-

Table 2: Strict 1,848-ID paired audit.

Metric	MoL-700	v0.43	Δ
Accuracy	0.802528	0.802528	0
Planning /100	73.1169	73.1169	0
MC accuracy	0.778966	0.778966	0
Coord. F1	0.146465	0.146465	0
Match /100	30.7513	30.7513	0
Language	0.475440	0.480405	+0.004965
Final	0.609563	0.610557	+0.000993

gate improves. This metric disagreement is a limitation, not omitted evidence.

4 Discussion

The result supports a specific interpretation of Graph-VQA: anchor perception and downstream reasoning should not necessarily share one adapter. Graph-A learns a better frame-level anchor representation, while MoL retains stronger task-specialized reasoning. The fixed composition avoids the negative transfer observed when Graph-SFT alone was used for every node.

This is not a parameter-level reproduction of InternVL4Drive and the scores are not directly comparable. InternVL used 64 A100 GPUs, full-parameter tuning, an official GPT judge and hidden leaderboard data. We use retained 7B LoRA models, a local scene-isolated split and a DeepSeek proxy. What is reproduced is the experimental principle of complementary specialization and ensembling.

5 Next Steps

We will evaluate three remaining transfers with equal-budget controls: (1) explicit six-camera topology prompts while retaining six independent images; (2) a labeled 2×3 mosaic plus global thumbnail with the same total visual token budget; and (3) train-only point-to-box auxiliary grounding followed by inverse conversion to official center coordinates. The promoted v0.43 output should also be exported to official validation format and scored on the hidden server before any leaderboard claim.

References

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